Filbin 2018 DIPG (H3-K27M) — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity — GO

5/6 samples (≥184 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached components); per-neighbourhood iLISI 3.66/5 at θ=10. Every lobe is multi-patient.

Gate 2 — module geography (non-branching) — GO

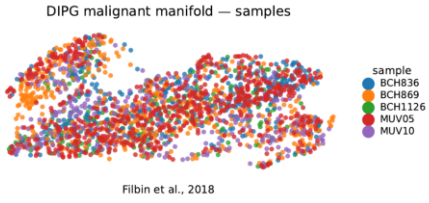
Gast & pSom co-vary (smoothed r=+0.79) rather than splitting into arms; Crest is the inverse pole (r=−0.89 vs Gast, −0.57 vs pSom), rising to a terminal readout (ρ=+0.91; crest-high decile mean pt=1.00).

Gate 3 — hub cross-repression — NA

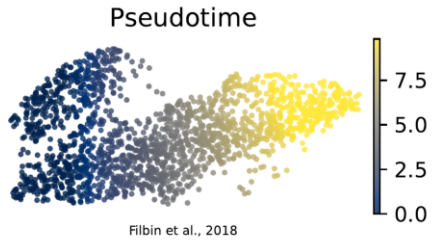
Non-branching: a single Gast/pSom→Crest axis with no Gast↔pSom bifurcation and no Mixed-type convergence hub, so there is no branch pair / hub population to test for cross-repression.

STEP FIGURES

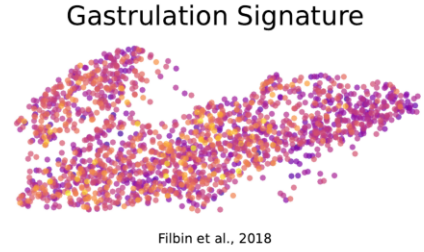
Gate 1: sample integration



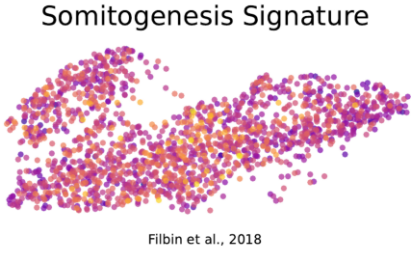
Trajectory / pseudotime (root: Gast/pSom-high)



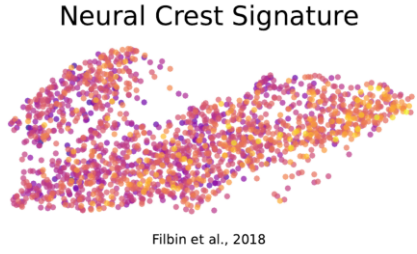
Gastrulation (raw)



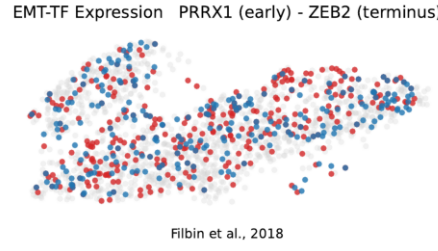
Somitogenesis (raw)



Neural crest (raw)



EMT-TFs: PRRX1 (root) · ZEB2 (late)



Dataset.
Pediatric H3-K27M mutant glioma / DIPG (Filbin et al., 2018), SmartSeq2 — 7 patients / 2,587 cells (master scope). Malignant only: 2,259 loaded; 2,118 across 5 samples (≥184-cell threshold, MUV01 dropped) enter preprocessing; Gate 1 (mixing/continuity) is judged on the whole malignant compartment, and the trajectory is scoped to the non-cycling developmental compartment (1,746 cells; see Verdict).

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Filbin's DIPG malignant manifold is a single monotone Gast/pSom→terminal-Crest developmental gradient, not Puram's hub-and-spoke bifurcation, so there is no hub to test at Gate 3. The tumour's genuine OPC-like→OC/AC-like hierarchy (opc↔ac r=−0.79) is a separate axis orthogonal to the Gast/pSom/Crest modules, and a PPT tree fit does not separate the modules. Trajectory scoped to the non-cycling developmental compartment: the 372 cycling cells form a coherent sub-region coincident with the Crest terminus, and on the whole manifold pseudotime tracks proliferation (cyc↔pt ρ=+0.53). A leave-out test shows the reference module geography survives their removal (Gast↔pSom +0.56→+0.79), so they are a proliferation confound, not part of the axis; scoping halves the confound (scoped cyc↔pt ρ=+0.34) and sharpens the gradient. EMT-TFs PRRX1 (root/early) → ZEB2 (terminus).

Parameter choices

Min cells / sample:
≥184 (geom-mean gap 241→141; keeps 5/6)

Trajectory scope:
non-cycling developmental compartment (1,746 / 2,118)

Root node:
curve tip 2 — Gast/pSom-high, Crest-low

Integration:
Harmony on sample, θ=10; PCA(50), n_pcs=20; spread=1.3

Trajectory:
scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; fold-free (SA 2.99)

Geometry:
Linear-pSom/Crest · EMT-TFs PRRX1→ZEB2